

Magnetic Fields — Paper D

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Write in sentences — a list of single words scores badly on this topic.
- Every field line you draw must carry an **arrowhead**. A line without one does not score.
- Where a question asks for a number, that number must carry a **unit**.
- The mark for each question is shown in square brackets on the right.
- No rules and no reminders are given on this paper.
- Several questions describe experiments you may not have seen before — apply the ideas you already have.

SECTION 1 – ARRANGEMENTS AND EVIDENCE

1. Two bar magnets lie in a line on a bench.

- (a)** Describe how they must be arranged so that a neutral point exists between them. [2]

- (b)** State what a neutral point is. [1]

- (c)** One of the magnets is replaced by a weaker one, still arranged the same way. [2]
State how the position of the neutral point changes, and explain why.

2. Kabir has two steel bars that look identical. One is a magnet; the other is not. He has nothing else — no compass, no iron filings, no second magnet.

- (a) Describe a test using only the two bars that tells him which one is the magnet, [3]
and state what he would observe in each case.

- (b) Explain why the test works. [2]

SECTION 2 – DIAGRAMS AND MATERIALS

3. A student hands in a field diagram for a bar magnet with four faults: the lines **stop in mid-air**; they are **evenly spaced everywhere**; the arrows point **into** the north pole; and two of the lines **touch** near a pole. For each fault, state what is wrong and what should have been drawn. [4]

4. Compass needles are made from steel. The cores of electromagnets are made from soft iron.

- (a) Explain why each material is the right choice for its job. [2]

- (b)** Bar magnets left loose in a drawer gradually get weaker. State one way magnets should be stored, and explain how it helps. [2]

SECTION 3 – JUDGEMENT

5. A hiker uses a compass.

- (a)** Explain why saying “the compass points to the North Pole” is loosely worded. [2]
Use the words *geographic* and *magnetic* in your answer.

- (b)** Standing next to a large steel gate, she finds the compass gives a different reading. Explain why. [2]

6. A bar magnet is dipped into a tray of iron filings and lifted out.

- (a)** Describe where the filings collect and explain why. [2]

- (b)** The magnet is then snapped in half. State what you now have, and explain. [2]

7. Another student has written two answers, and both are wrong. [4]
- (i) "The paper clip stuck to the magnet, so the paper clip must be a magnet as well."
- (ii) "I drew the field lines running from the south pole to the north pole outside the magnet, and I made them all the same distance apart."
- For each one, explain the mistake and give the correct version.

End of paper. Before you hand it in, go back over every diagram and check that each field line carries an arrowhead pointing from north to south.